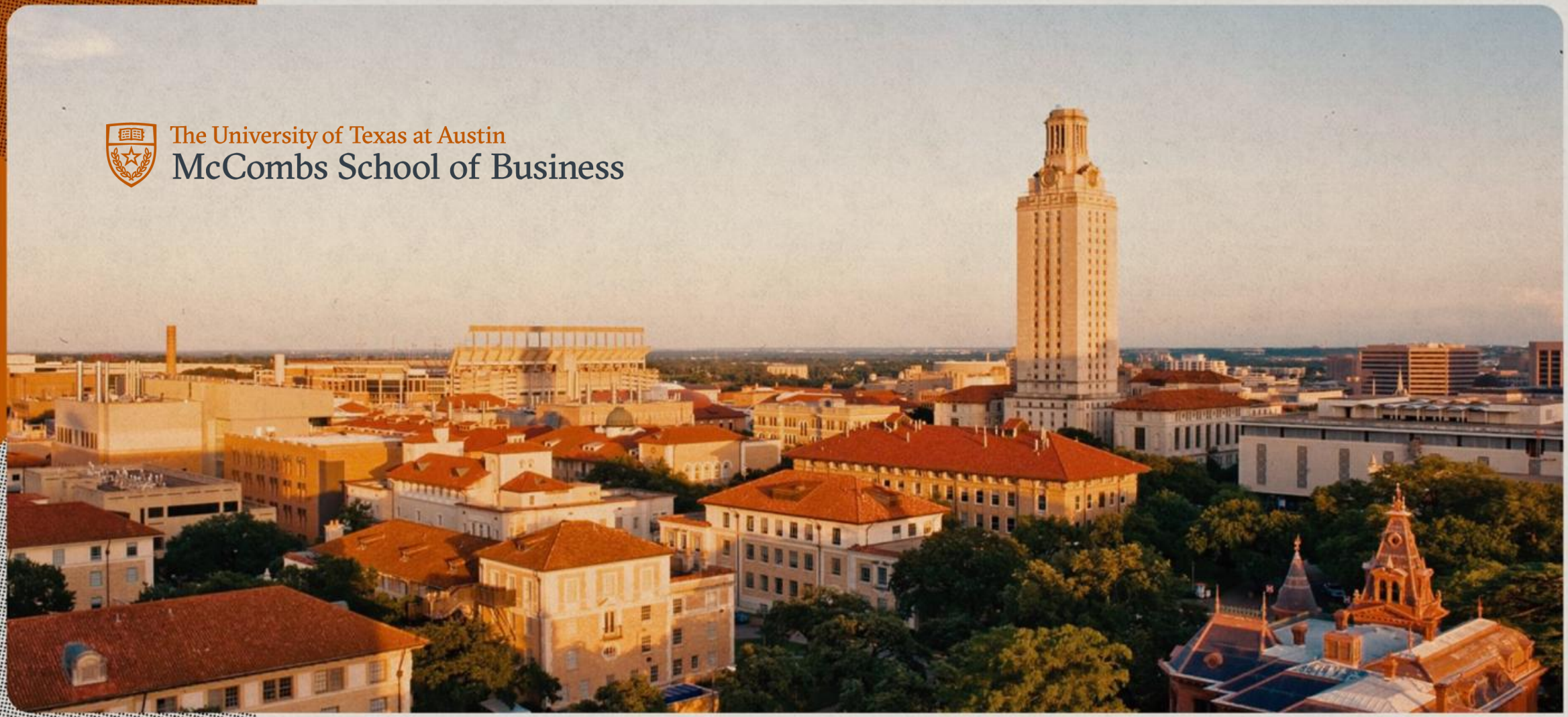

D13: apriori() Association Rules

Association Rules from 40,000 feet

At a general level, Association Rules (AR) say that certain items are likely to occur together with other items. The purpose of AR Learning (ARL) is to discover those associations in data and to evaluate their strengths.

The canonical setting for ARL is transactions data. ARL is much more widely applicable than just to sales data, but it is useful to learn about ARL in the transactions context, as the sales context provides a concrete framework to fix ideas and concepts.

Basic Ideas of ARL

Consider the following rule from an analysis of grocery store point-of-sales data:

{bread, butter} => {milk}

This rule says that the purchase of milk is associated with the purchase of bread and butter. {bread, butter} and {milk} are called **itemsets**, which are simply items that may occur together, including singly. An **association rule** is an implication, written **A => B**, in which A and B are itemsets of single or multiple items.

ARL involves two main steps:

1. Frequent itemset generation – find those itemsets that occur often in the data
2. Rule generation – finding itemsets that are related in various metrics

Basic Ideas of ARL

Example: Consider a dataset with 100 transactions, of which …

A = {bread} occurs in 40 (possibly along with other items)

B = {butter} occurs in 30 (possibly along with other items)

C = {bread, butter} occurs in 20 (possibly along with other items)

and neither occurs in 50 transactions *[TA Q: how do you get 50 in neither?]*

So A, B, C are like overlapping circles in a Venn diagram.

Basic Ideas of ARL

The **support** of an itemset X is the proportion of cases in the data in which X occurs: **Support(X) = # times X occurs / # total data**.

In the Example …

Support(A) = Support({bread}) = 40 / 100 = 0.40

Support(B) = Support({butter}) = 30 / 100 = 0.30

Support(C) = Support({bread, butter}) = 20 / 100 = 0.20

If we think of the transactions as being “equally likely”, then the support of an itemset is the unconditional probability of the itemset: P(X).

Basic Ideas of ARL

The **confidence** of an AR X => Y is the proportion of cases when Y occurs among the cases in which X already occurs – so it is the count of Y within the count of X:

Confidence(X=>Y) = # times both X and Y occur / # times X occurs

In the Example …

Conf(A=>B) = Conf({bread}=>{butter}) = 20 / 40 = 0.50

Conf(B=>A) = Conf({butter}=>{bread}) = 20 / 30 = 0.67

Conf(C=>A) = Conf({bread, butter}=>{bread} = 20 / 20 = 1.00

In a probability interpretation, the confidence of X=>Y is the conditional probability of Y given X: P(Y | X).

Basic Ideas of ARL

The **lift** of an AR X => Y is **Lift(X=>Y) = Conf(X=>Y) / Support(Y)**

Lift(X=>Y) measures the relative boost ("lift") in the probability that Y will occur, given that X occurs, relative to not knowing that X occurs. So lift measures the improvement in the odds for Y that results from knowing that X is associated with Y. Lift could be less than 1, in which case knowing X occurs reduces the odds of Y.

In the Example …

Lift(A=>B) = Conf({bread}=>{butter}) / Supp({butter} = 0.50 / 0.30 = 1.67

Lift(B=>A) = Conf({butter}=>{bread}) / Supp({bread}) = 0.67 / 0.40 = 1.675

Lift(C=>A) = Conf({bread, butter}=>{bread}) / Supp({bread}) = 1 / 0.40 = 2.50

How apriori() Works

`apriori()` iteratively reduces the number of candidate itemsets by applying the **Apriori property**: *If an itemset is frequent, all of its subsets must also be frequent*.

Apriori uses a breadth-first search strategy and a hash tree structure to count candidate itemsets efficiently.

To use the Apriori algorithm, the data must be in the form of a **transaction dataset**. Each transaction is a collection of items, and the dataset is a list of such transactions. In R, this is typically represented as a `transactions` object from the `arules` package. (See illustration of conversion of Excel data into proper format with the “test_apriori_data.xlsx” example.)

How apriori() Works

Typical parameter values specified in apriori():

Support: Minimum support threshold to consider an itemset as frequent.

Confidence: Minimum confidence threshold to consider a rule as strong.

Lift: Measure of the strength of a rule over random co-occurrence.

Output can be visualized by a variety of plots, available in the `arulesViz` package.

Output plots in Association Rule Learning can include:

Scatter Plots: Showing the relationship between support, confidence, and lift of the rules.

Matrix Plots: Visualizing the support of itemsets.

Graph-Based Plots: Representing the rules as a network of items.

Major Applications of ARL

Market Basket Analysis: Identifying products that are frequently bought together.

Recommendation Systems: Suggesting items to users based on their purchase history.

Fraud Detection: Detecting unusual patterns that may indicate fraudulent activity.

Healthcare: Discovering associations between symptoms and diseases.

Advantages of ARL

Simplicity: Easy to understand and implement.

Interpretability: The rules generated are easy to interpret and can provide actionable insights.

Scalability: Efficient algorithms like A Priori can handle large datasets.

Disadvantages of ARL

Computational Complexity: Generating frequent itemsets can be computationally expensive, especially for large datasets.

Redundancy: Many rules generated can be redundant or not interesting.

Parameter Sensitivity: The results can be sensitive to the choice of support and confidence thresholds.